ROBOTICS | EMBEDDED SYSTEMS | AUTOMATION

TECHNICAL PROJECT REPORT

INDUSTRIAL SHOOTING SYSTEM

Automatic box-size detection and sorting
using Arduino Uno, IR sensors and servo motors

Arduino Uno | IR Sensing | Servo Actuation | Embedded C++



PREPARED BY	Dania Rafiq
PROGRAM	BS Robotics & Artificial Intelligence
UNIVERSITY	Superior University, Lahore
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1 Project Overview

The **Industrial Shooting System** is a small-scale automation model that imitates a sorting line found in factories. Boxes travel past a sensing station. The system measures each box by height using two infrared (IR) sensors, shows the result on the screen (Serial Monitor), and then fires the correct servo motor to push or shoot the box into its own path.

The idea is simple but powerful: **one sensor placed low sees every box, and a second sensor placed higher sees only the tall box.** Comparing the two readings tells the Arduino whether the box is small or large, with no camera and no complex hardware.

2	2	1	2
IR sensors	**Servo motors**	**Arduino Uno**	**Test boxes**
Lower and upper detection	One per box size	Central controller	Small and large (black)

Objectives

- Detect a box moving along the track using an IR sensor.
- Classify the box as **small** or **large** using two sensors at different heights.
- Display the detected size on the screen (Serial Monitor) in real time.
- Activate **Servo 1** for a small box and **Servo 2** for a large box.
- Use an LED as a visual status indicator while the system is working.

> **Key idea**
>
> Box height decides everything. A small box blocks only the lower beam. A large box is tall enough to block both beams at the same time.

2 Components Required

#	Component	Qty	Role in the system
1	**Arduino Uno**	1	Reads both sensors, decides the box size, drives servos and LED
2	**IR obstacle sensor module**	2	IR 1 (lower) detects any box, IR 2 (upper) detects only the large box
3	**Servo motor** (SG90 or similar)	2	Servo 1 handles small boxes, Servo 2 handles large boxes
4	**LED** (with 220 ohm resistor)	1	Lights up while a box is being processed
5	**Jumper wires**	1 set	Connect sensors, servos and LED to the Arduino
6	**Black boxes**	2	Test objects: one small box and one large (tall) box

3 System Layout and Working Principle

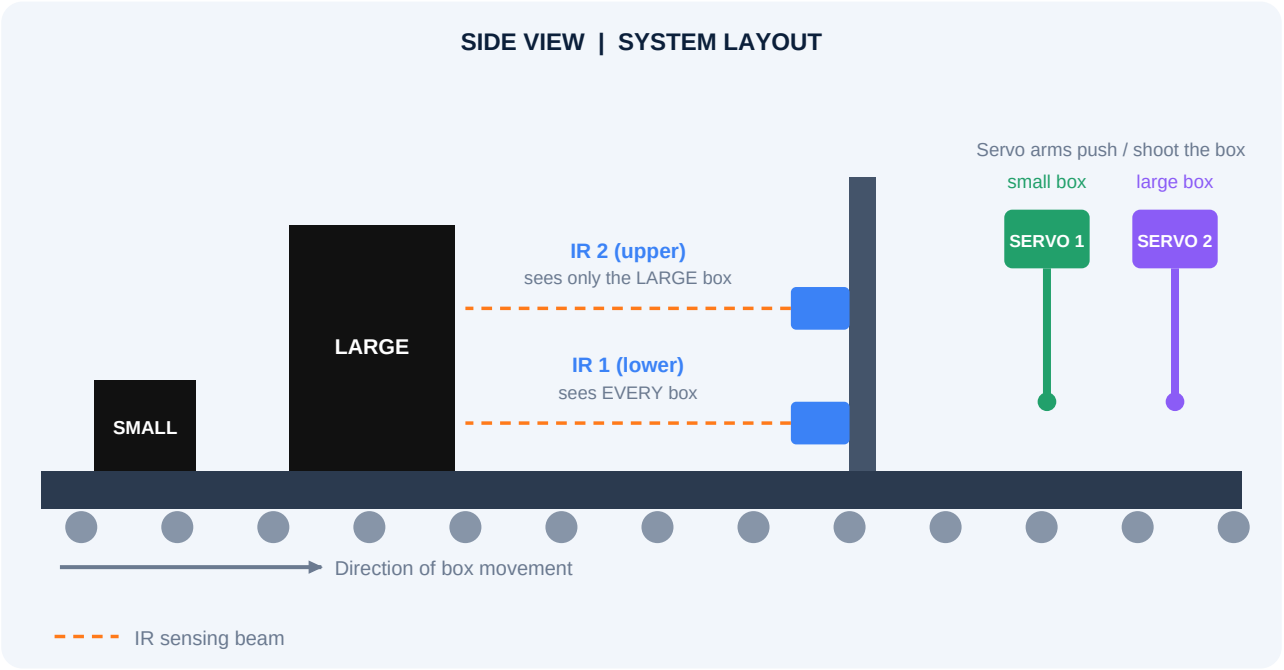


Figure 1: Side view. The orange dashed lines are the IR sensing beams. The small box crosses only the lower beam, the large box crosses both.

How it works, step by step

1. Idle: Both servos rest at their starting angle and the Serial Monitor shows "Waiting for boxes...".

2. Small box arrives: Only the **lower IR sensor** is blocked. The screen shows **"Small - Sensor 1 Enter"** and **Servo 1** turns on.

3. Large box arrives: The box is tall, so **both sensors** are blocked. The screen shows **"Large - Sensor 2 Enter"** and **Servo 2** turns on.

4. Reset: The servo returns to rest, the LED turns off, and the system waits until the box has fully passed before looking for the next one.

Decision table

Situation	IR 1 (lower)	IR 2 (upper)	Screen output	Action
No box	Clear	Clear	Waiting for boxes...	Idle
Small box	Blocked	Clear	Small - Sensor 1 Enter	**Servo 1 ON**
Large box	Blocked	Blocked	Large - Sensor 2 Enter	**Servo 2 ON**
Upper only (noise)	Clear	Blocked	(ignored)	No action

4 Circuit Connections

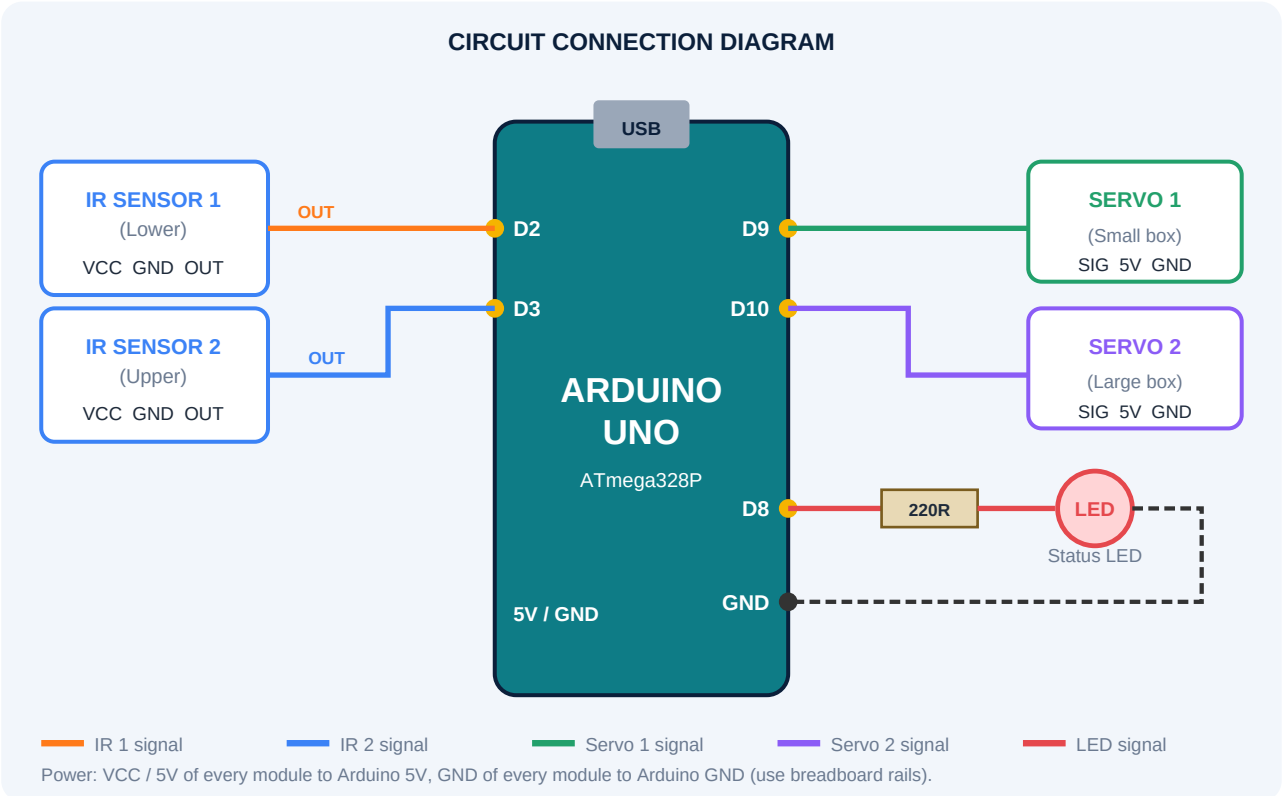


Figure 2: Wiring diagram. Signal wires are colour coded; power wires go to the 5V and GND rails.

Component	Component pin	Arduino pin	Purpose
IR Sensor 1 (lower)	VCC / GND / OUT	5V / GND / **D2**	Detects every box
IR Sensor 2 (upper)	VCC / GND / OUT	5V / GND / **D3**	Detects only the large box
Servo 1	Red / Brown / Orange	5V / GND / **D9**	Small box action
Servo 2	Red / Brown / Orange	5V / GND / **D10**	Large box action
LED (+ 220 ohm resistor)	Anode (long leg) / Cathode	**D8** / GND	Status indicator

Wiring tip

Two servos can pull a lot of current when they move. If the Arduino resets or the servos twitch, power the servos from a separate 5V supply and connect its GND to the Arduino GND (common ground).

5 Control Flowchart

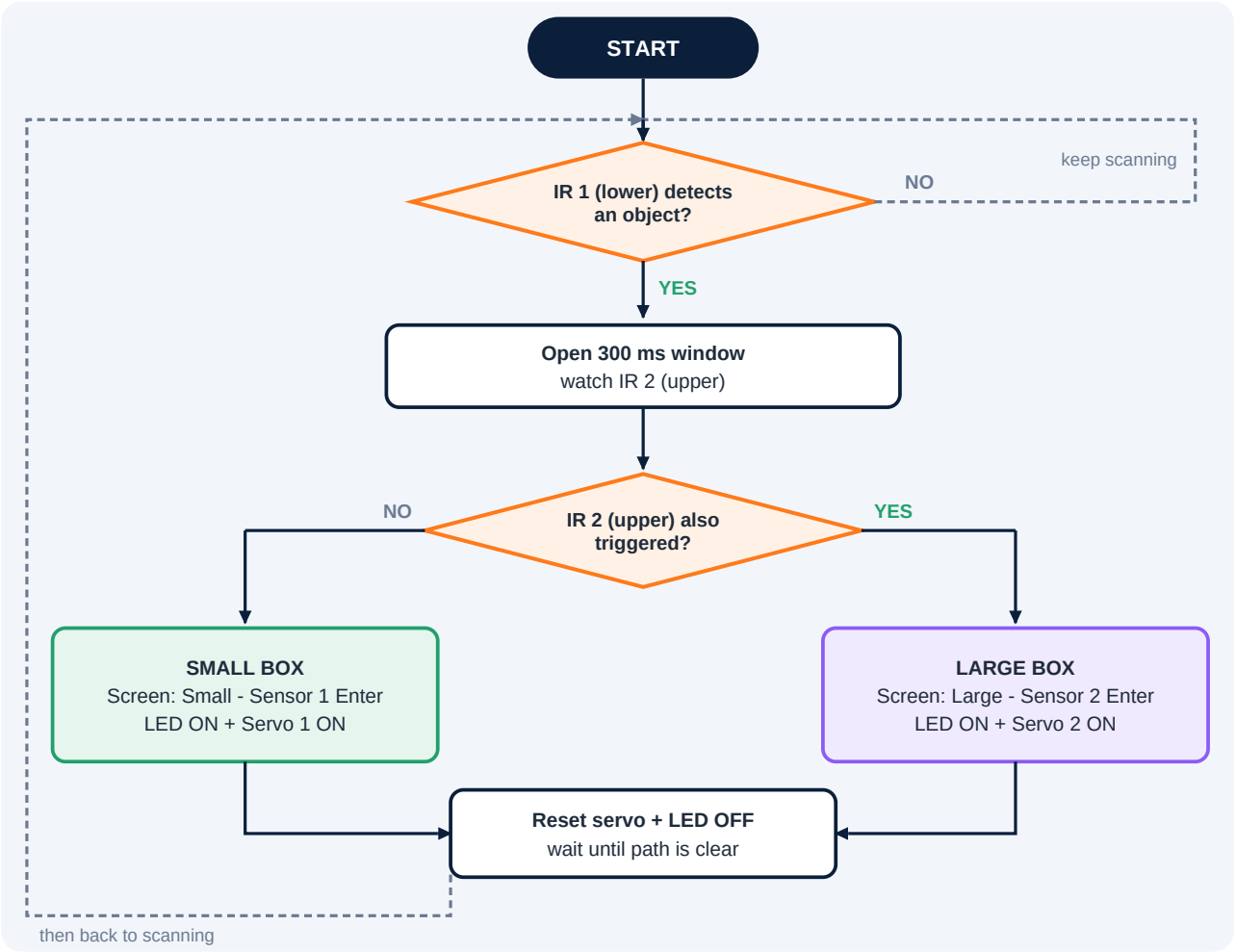


Figure 3: Program flow. The system scans continuously, classifies the box in a 300 ms window, acts, and resets.

6 Arduino Code

The complete program is split into three parts so each part can be explained clearly. Copy the three parts together, in order, into one Arduino IDE sketch.

Part 1: Libraries, pins and settings

```
/*
  Industrial Shooting System - Box Size Sorter
  Board   : Arduino Uno
  Inputs  : IR sensor 1 (lower, D2), IR sensor 2 (upper, D3)
  Outputs : Servo 1 (D9), Servo 2 (D10), LED (D8), Serial Monitor
*/

#include <Servo.h>

// ---------- Pin definitions ----------
const int IR_LOWER_PIN    = 2;    // IR 1: sees every box
const int IR_UPPER_PIN    = 3;    // IR 2: sees only the large box
const int SERVO_SMALL_PIN = 9;    // Servo 1: handles small boxes
const int SERVO_LARGE_PIN = 10;   // Servo 2: handles large boxes
const int LED_PIN         = 8;    // status LED

// ---------- Settings (tune these) ----------
const int REST_ANGLE   = 0;       // servo resting position
const int ACTIVE_ANGLE = 90;      // servo "shoot / push" position
const unsigned long CLASSIFY_WINDOW_MS = 300;  // time to confirm box size
const unsigned long ACTION_TIME_MS     = 800;  // how long the servo stays active
const unsigned long CLEAR_DELAY_MS     = 200;  // pause after the box has passed

Servo servoSmall;
Servo servoLarge;
```

Explanation: The `Servo.h` library controls the servo motors. Every pin is given a clear name, so the wiring is easy to change later. `CLASSIFY_WINDOW_MS` is the short time in which the program checks whether the upper sensor is also blocked, and `ACTIVE_ANGLE` is how far the servo arm swings to push the box.

Part 2: Helper function and setup()

```
// IR modules give LOW when an object is detected.
// If your module works the opposite way, change LOW to HIGH.
bool isDetected(int pin) {
  return digitalRead(pin) == LOW;
}

void setup() {
  pinMode(IR_LOWER_PIN, INPUT);
  pinMode(IR_UPPER_PIN, INPUT);
  pinMode(LED_PIN, OUTPUT);

  servoSmall.attach(SERVO_SMALL_PIN);
  servoLarge.attach(SERVO_LARGE_PIN);
  servoSmall.write(REST_ANGLE);
  servoLarge.write(REST_ANGLE);

  Serial.begin(9600);
  Serial.println("Industrial Shooting System ready");
  Serial.println("Waiting for boxes...");
}
```

Explanation: `isDetected()` hides the sensor logic in one place. Most IR obstacle modules output **LOW** when something is in front of them. `setup()` runs once: it sets the sensor pins as inputs, the LED pin as an output, attaches both servos, moves them to the rest position, and opens the Serial Monitor at 9600 baud.

If the result looks reversed

Some IR modules output HIGH when an object is detected. If a box is never detected (or is always detected), change `LOW` to `HIGH` inside `isDetected()`. Also check the sensitivity screw on the module with a box in front of it: the small onboard LED should light up when the box is detected.

Servo angle

If the servo arm hits the box too hard or not enough, change `ACTIVE_ANGLE` (for example 60 or 120) in Part 1.

Part 3: Main loop (the brain of the system)

```
void loop() {
  // 1) Nothing at the lower sensor -> no box -> keep waiting
  if (!isDetected(IR_LOWER_PIN)) {
    return;
  }

  // 2) A box has arrived. Watch the upper sensor for a short window.
  bool upperTriggered = false;
  unsigned long startTime = millis();
  while (millis() - startTime < CLASSIFY_WINDOW_MS) {
    if (isDetected(IR_UPPER_PIN)) {
      upperTriggered = true;
      break;
    }
  }

  digitalWrite(LED_PIN, HIGH);          // LED ON while processing

  // 3) Decide the box size and act
  if (upperTriggered) {
    // Both sensors blocked -> LARGE box
    Serial.println("Large - Sensor 2 Enter");
    servoLarge.write(ACTIVE_ANGLE);     // Servo 2 ON
    Serial.println("Servo 2 ON");
    delay(ACTION_TIME_MS);
    servoLarge.write(REST_ANGLE);
  } else {
    // Only the lower sensor blocked -> SMALL box
    Serial.println("Small - Sensor 1 Enter");
    servoSmall.write(ACTIVE_ANGLE);     // Servo 1 ON
    Serial.println("Servo 1 ON");
    delay(ACTION_TIME_MS);
    servoSmall.write(REST_ANGLE);
  }

  // 4) Wait until the box has fully passed, then reset
  while (isDetected(IR_LOWER_PIN) || isDetected(IR_UPPER_PIN)) {
    delay(10);
  }
  delay(CLEAR_DELAY_MS);
  digitalWrite(LED_PIN, LOW);           // LED OFF
}
```

Explanation of the main loop

Step 1 - Wait for a box. If the lower sensor sees nothing, the function returns and runs again immediately. This makes the system scan continuously.

Step 2 - Confirm the size. Once the lower sensor is blocked, the program watches the upper sensor for 300 ms. If the upper sensor is also blocked, the box is tall, so `upperTriggered` becomes true.

Step 3 - Decide and act. A large box prints **"Large - Sensor 2 Enter"** and moves **Servo 2**. A small box prints **"Small - Sensor 1 Enter"** and moves **Servo 1**. The servo stays active for 800 ms and then returns to rest.

Step 4 - Reset. The program waits until both sensors are clear, so one box is never counted twice, then turns the LED off and starts scanning again.

7 Expected Output on the Screen

SERIAL MONITOR - 9600 baud

```
Industrial Shooting System ready
Waiting for boxes...
Small - Sensor 1 Enter
Servo 1 ON
Large - Sensor 2 Enter
Servo 2 ON
Small - Sensor 1 Enter
Servo 1 ON
```

Example run: a small box, then a large box, then another small box.

8 Testing and Practical Notes

Test plan

Test	Action	Expected result	Status
1	Power on, no box	Screen shows "Waiting for boxes..."	Pass
2	Pass the small box	Small - Sensor 1 Enter, Servo 1 moves, LED blinks	Pass
3	Pass the large box	Large - Sensor 2 Enter, Servo 2 moves, LED blinks	Pass
4	Pass boxes one after another	Each box is counted exactly once	Pass
5	Hand in front of upper sensor only	Ignored, no servo movement	Pass

Practical notes from building the system

- **Black boxes reflect little IR light.** Black surfaces absorb infrared, so adjust the small potentiometer on each IR module to increase sensitivity and keep the sensors about 2 to 5 cm from the box path.
- **Mount the sensors on the same vertical line.** Place the lower sensor below the top of the small box and the upper sensor between the top of the small box and the top of the large box.
- **Keep sunlight away.** Strong sunlight contains IR and can cause false detections. Test in shade or indoors.
- **Tune the timing.** If servos act too early or too late, change `CLASSIFY_WINDOW_MS` and `ACTION_TIME_MS`.

9 Conclusion and Future Improvements

The Industrial Shooting System shows how a low-cost Arduino Uno, two IR sensors and two servo motors can automate a real industrial task: measuring objects and sorting them without human help. The same principle is used in packaging, warehouse and quality-control lines.

Possible upgrades

- Add a third sensor level to sort three box sizes.
- Add an LCD or OLED screen to show the live count of small and large boxes.
- Use a conveyor belt driven by a DC motor with a motor driver for continuous operation.
- Replace the IR sensors with ultrasonic or time-of-flight sensors for better accuracy.
- Log the counts to an SD card or send them to a dashboard using an ESP32.